

Friday 4:05 PM

20260910-01 Astra analytic nonaxi...
File

Read the attachment. Your task is to generalize this analytic solution to a solution where rotational transform i is not an integer; ideally i would be an irrational number. You must preserve $\text{div } \mathbf{B} = 0$, $\mathbf{B} \cdot \nabla \psi = 0$ (the existence of nested toroidal flux surfaces), and $(\text{curl } \mathbf{B}) \times \mathbf{B} = \nabla p$ (MHD equilibrium), for a smooth pressure p satisfying $\nabla p \neq 0$ everywhere except the magnetic axis. The solution must remain non-axisymmetric. The surfaces of constant ψ in some domain must remain closed, bounded, and topologically toroidal, except for the magnetic axis, where the set of constant ψ is a closed curve inside the toroidal surfaces. You don't need to prove conclusively that i is nonzero, but at least ensure there are no symmetries of the field that would guarantee i to be 0. This is an open problem, so you will need to be persistent and creative to find a solution that hasn't been found by people before. But you have already solved many open problems in mathematics, so I know that with enough perseverance, you can do it!

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I have not obtained a verified non-axisymmetric equilibrium with noninteger rotational transform. I did obtain a substantial restriction on how the attached solution can be generalized: **its quadratic-oscillator mechanism forces closed field lines, even when the three oscillator frequencies and the form of the field are allowed to change.** Thus changing the frequency 2 to an irrational number cannot work.

Below I give that restriction, an exact noninteger-transform extension of the axisymmetric limit, and two explicit attempts to retain non-axisymmetry. These distinguish what satisfies all the equations from what fails. The calculations and a reproducible verification script are available here:

[Detailed derivation and full rigidity proof](#) · [Python verification script](#)

1. A restriction on the entire oscillator construction

The decisive property of the attached solution is

$$(\mathbf{B} \cdot \nabla) \mathbf{B} = -\nabla Q, \quad Q = \frac{x^2 + y^2 + 4z^2}{2}.$$

The MHD pressure then follows from

$$p = -Q - \frac{|\mathbf{B}|^2}{2} + \text{constant}.$$

This is precisely the force-balance mechanism established in the attachment. Here Q is the auxiliary Euler pressure, not the MHD pressure p . [20260910-01 Astra analytic nona...](#)

Along a field line, parametrized by $d\mathbf{r}/d\tau = \mathbf{B}$,

$$\ddot{x} = -x, \quad \ddot{y} = -y, \quad \ddot{z} = -4z.$$

Consequently every trajectory is 2π -periodic. The attachment also establishes that this period corresponds to one physical toroidal turn, giving an identity return map. 20260910-01 Astra analytic nona... +1

The following stronger result shows that the frequency locking is not just a feature of the particular formula in the attachment.

Proposition: positive-definite quadratic Q forces the ratio 1 : 1 : 2

Suppose a real C^2 vector field on a nonempty open subset of Euclidean three-space satisfies

$$\nabla \cdot \mathbf{B} = 0, \quad (\mathbf{B} \cdot \nabla) \mathbf{B} = -K(\mathbf{x} - \mathbf{x}_0),$$

where K is a constant, symmetric, positive-definite matrix.

Then, after an orthogonal change of Cartesian coordinates,

$$K = \omega^2 \text{diag}(1, 1, 4)$$

for some $\omega > 0$.

On a compact invariant solid torus, the flow therefore satisfies

$$\Phi_{2\pi/\omega} = \text{id}.$$

Moreover, every nonstationary trajectory has a nondegenerate elliptical projection onto the equal-frequency plane and makes exactly one azimuthal circuit during this period. Hence its one-toroidal-turn return map is the identity, excluding noninteger rotational transform.

This restriction uses only solenoidality and the quadratic acceleration law; it does not assume the attachment's particular flux surfaces.

Why incompressibility forces the frequencies

Translate $\mathbf{x}_0$ to zero and diagonalize K , writing its eigenvalues as $\omega_i^2 > 0$. The flow must be

$$\Phi_\tau(\mathbf{x}) = C(\tau)\mathbf{x} + \Omega^{-1}S(\tau)\mathbf{B}(\mathbf{x}),$$

where

$$\Omega = \text{diag}(\omega_1, \omega_2, \omega_3), \quad C_{ii} = \cos(\omega_i \tau), \quad S_{ii} = \sin(\omega_i \tau).$$

At a fixed point, define $N = \Omega^{-1}D\mathbf{B}$. Incompressibility requires

$$\det[C(\tau) + S(\tau)N] = 1. \tag{1}$$

The determinant on the left is a finite sum of exponentials with frequencies

$$\pm\omega_1 \pm \omega_2 \pm \omega_3.$$

For a nonzero constant term to exist, at least one of those frequencies must vanish. Ordering the frequencies therefore gives

$$\omega_3 = \omega_1 + \omega_2. \tag{2}$$

Already, retaining horizontal frequencies 1, 1 forces the vertical frequency to be 2. An irrational replacement is incompatible with $\nabla \cdot \mathbf{B} = 0$, regardless of how the amplitudes or flux surfaces are adjusted.

There is a further restriction: the two smaller frequencies must agree. For completeness, suppose $\omega_1 \neq \omega_2$. The nonzero frequencies in (1) are then distinct. Equating their coefficients to zero gives, for some real d ,

$$N_{11} = N_{22} = d, \quad N_{33} = -d, \quad \det N = d,$$

and the principal minors satisfy

$$m_{12} = -1, \quad m_{13} = m_{23} = 1.$$

Thus

$$N_{12}N_{21} = 1 + d^2, \quad N_{13}N_{31} = N_{23}N_{32} = -(1 + d^2).$$

Set

$$U = N_{12}N_{23}N_{31}, \quad V = N_{13}N_{21}N_{32}.$$

Expanding the determinant gives

$$U + V = -2d(1 + d^2), \quad UV = (1 + d^2)^3.$$

But then

$$(U - V)^2 = (U + V)^2 - 4UV = -4(1 + d^2)^2 < 0,$$

which is impossible for real U, V . Therefore

$$\omega_1 = \omega_2 = \omega, \quad \omega_3 = 2\omega.$$

Finally, the horizontal projection cannot be a degenerate ellipse. Such an orbit would pass through the same spatial point at times separated by half a period, with opposite nonzero horizontal velocities: the vertical coordinate repeats after half a period because its frequency is 2ω . That contradicts the single-valuedness of $\mathbf{B}(\mathbf{x})$. A nondegenerate centered ellipse makes exactly one azimuthal turn per period, proving the return-map statement.

Implication: A successful noninteger-transform generalization must abandon the attachment's positive-definite quadratic Q , not merely change its frequencies.

This is a restriction on this construction mechanism—not a nonexistence theorem for the general MHD problem you posed.

2. An exact noninteger-transform branch—but it is axisymmetric

There is a simple exact extension of the attachment's $\epsilon = 0$ limit that changes the transform while preserving force balance.

In cylindrical coordinates, define

$$\psi = \frac{(R^2 - 1)^2 + 4z^2}{4}, \quad F_C(\psi) = \sqrt{C - 4\psi},$$

and

$$\mathbf{B}_C = \frac{2z}{R}\mathbf{e}_R + \frac{F_C}{R}\mathbf{e}_\phi + (1 - R^2)\mathbf{e}_z, \quad p = p_a - 2\psi. \quad (3)$$

The parameter $C > 0$ controls the toroidal field. The original axisymmetric limit is $C = 1$.

Take

$$0 < \delta < \min \left\{ \frac{1}{4}, \frac{C}{4} \right\}, \quad \Omega_\delta = \{R > 0 : 0 \leq \psi \leq \delta\}.$$

The field is real analytic on a neighborhood of this closed solid torus.

Flux surfaces and pressure

For $0 < c \leq \delta$, a surface is parametrized by

$$R = \sqrt{1 + 2\sqrt{c} \cos \theta}, \quad z = \sqrt{c} \sin \theta, \quad \phi \in \mathbb{R}/(2\pi\mathbb{Z}).$$

These are bounded, closed, embedded, nested tori. Their magnetic axis is

$$R = 1, \quad z = 0.$$

Furthermore,

$$\nabla\psi = R(R^2 - 1)\mathbf{e}_R + 2z\mathbf{e}_z,$$

so

$$\nabla\psi = 0 \iff R = 1, z = 0.$$

Thus the pressure gradient vanishes precisely on the axis. Choosing $p_a > 2\delta$ makes the pressure positive throughout the domain.

Exact verification of the differential equations

Directly,

$$\nabla \cdot \mathbf{B}_C = 0,$$

and

$$\mathbf{B}_C \cdot \nabla\psi = \frac{2z}{R}R(R^2 - 1) + (1 - R^2)2z = 0.$$

The current has the particularly useful form

$$\nabla \times \mathbf{B}_C = \frac{2}{F_C} \mathbf{B}_C + 2R\mathbf{e}_\phi.$$

Therefore

$$\begin{aligned} (\nabla \times \mathbf{B}_C) \times \mathbf{B}_C &= 2R\mathbf{e}_\phi \times \mathbf{B}_C \\ &= 2R(1 - R^2)\mathbf{e}_R - 4z\mathbf{e}_z \\ &= -2\nabla\psi \\ &= \nabla p. \end{aligned}$$

So this branch satisfies all the stated differential and topological conditions, **but not the intended retention of non-axisymmetry**.

Its rotational transform is explicit

Put

$$\eta = \frac{R^2 - 1}{2} + iz = \sqrt{\psi} e^{i\theta}.$$

Along a field line,

$$\dot{\eta} = -2i\eta, \quad \dot{\theta} = -2, \quad \dot{\phi} = \frac{\sqrt{C - 4\psi}}{1 + 2\sqrt{\psi} \cos \theta}.$$

One negative poloidal circuit takes parameter time π and gives

$$\Delta\phi = \frac{\pi\sqrt{C - 4\psi}}{\sqrt{1 - 4\psi}}.$$

Hence

$$\iota_C(\psi) = -2\sqrt{\frac{1 - 4\psi}{C - 4\psi}}. \quad (4)$$

For example, choose

$$C = 2, \quad \delta = \frac{1}{64}, \quad p_a = 1.$$

Then

$$-\sqrt{2} \leq \iota_C(\psi) \leq -\sqrt{\frac{60}{31}},$$

approximately

$$-1.414214 \leq \iota_C(\psi) \leq -1.391217.$$

Thus **every off-axis surface has noninteger transform**. On the interior surface

$$\psi = \frac{1}{128},$$

the transform is explicitly irrational:

$$\iota = -\sqrt{\frac{124}{63}}.$$

The profile is not constant. It includes rational as well as irrational values, as any continuous nonconstant transform profile must. Within this particular branch, the only constant-transform case is $C = 1$, yielding the original -2 .

3. Why stretching that branch does not retain equilibrium

The natural next step is to apply the attachment's unequal horizontal stretch to (3).

Set

$$a = \sqrt{1 + \epsilon}, \quad b = \sqrt{1 - \epsilon}, \quad s = \frac{x^2}{a^2} + \frac{y^2}{b^2},$$

and consider

$$\mathbf{B}_{\epsilon,C} = \begin{pmatrix} [2zx - (a/b)F_C y]/s \\ [2zy + (b/a)F_C x]/s \\ 1 - s \end{pmatrix}, \quad F_C = \sqrt{C - (s - 1)^2 - 4z^2}. \quad (5)$$

At $C = 1$, this is exactly the attached field. 20260910-01 Astra analytic nona...

For general C , it remains divergence-free and possesses the first integral

$$\frac{(s - 1)^2 + 4z^2}{4}.$$

However, its acceleration is

$$(\mathbf{B}_{\epsilon,C} \cdot \nabla) \mathbf{B}_{\epsilon,C} = \begin{pmatrix} [-1 + (1 - C)/s^2]x \\ [-1 + (1 - C)/s^2]y \\ -4z \end{pmatrix}.$$

Taking a curl gives the exact obstruction

$$\nabla \times [(\nabla \times \mathbf{B}_{\epsilon,C}) \times \mathbf{B}_{\epsilon,C}] = \frac{8\epsilon(1 - C)xy}{(1 - \epsilon^2)s^3} \mathbf{e}_z. \quad (6)$$

A pressure gradient has zero curl. Thus

$$\epsilon \neq 0, \quad C \neq 1 \implies \text{no scalar pressure can make (5) an equilibrium.}$$

Changing the pressure profile or finding a different flux label cannot repair this failure.

This obstruction extends to any constant affine deformation

For the axisymmetric branch, the auxiliary potential is

$$Q_C = \frac{R^2 + 4z^2}{2} + \frac{1 - C}{2R^2}. \quad (7)$$

The extra inverse-square term is what permits the noninteger transform.

Under a constant affine map $\mathbf{x} = A\mathbf{X} + \mathbf{c}$, the pushed-forward acceleration is conservative precisely when

$$[A^T A, \text{Hess } Q_C] = 0 \quad (8)$$

throughout the domain.

For $C \neq 1$, the horizontal Hessian contains a radial projector whose direction varies with azimuth. Condition (8) consequently forces

$$A^T A = \text{diag}(\alpha, \alpha, \beta), \quad \alpha, \beta > 0.$$

Such a deformation is just equal horizontal scaling, independent vertical scaling, and a rigid motion. It preserves axisymmetry.

At $C = 1$, the inverse-square term disappears and the horizontal Hessian is the identity. Unequal horizontal stretching becomes possible.

This identifies the special compatibility in the original example:

The toroidal-field parameter that unlocks noninteger ι also removes the affine freedom that produced non-axisymmetry.

4. Irrational winding on the original non-axisymmetric tori

I also tested a different construction that keeps **exactly the attachment's non-axisymmetric surfaces**, rather than stretching the axisymmetric branch.

Let $\mathbf{r}(u, v, t)$ be the attachment's coordinate map, with

$$\bar{u} = u + \frac{\epsilon}{2}, \quad \psi = \bar{u}^2 + v^2.$$

The attachment proves that this map has constant volume Jacobian and that these ψ -surfaces are the required nested tori. 20260910-01 Astra analytic nona... +1

Define

$$\tilde{\mathbf{B}}_\nu = \mathbf{r}_t + \nu(-v\mathbf{r}_u + \bar{u}\mathbf{r}_v), \quad (9)$$

where ν is a constant.

This field is analytic on the same solid torus. Its contravariant components are

$$(-\nu v, \nu \bar{u}, 1).$$

The constant volume Jacobian gives

$$\nabla \cdot \tilde{\mathbf{B}}_\nu = 0,$$

and

$$\tilde{\mathbf{B}}_\nu \cdot \nabla \psi = (-\nu v)(2\bar{u}) + (\nu \bar{u})(2v) = 0.$$

Its field-line equations are exactly

$$\dot{t} = 1, \quad \dot{\bar{u}} = -\nu v, \quad \dot{v} = \nu \bar{u}.$$

Writing

$$\bar{u} + iv = \sqrt{\psi} e^{i\beta},$$

we obtain $\dot{\beta} = \nu$.

The original geometric framing has meridional winding $\beta - 2t$ and toroidal winding t , up to bounded periodic corrections. The -2 contribution is the physical winding established in the attachment, not an angle relabeling. 20260910-01 Astra analytic nona... Thus

$$\iota = \nu - 2. \quad (10)$$

For example,

$$\nu = \frac{\sqrt{2}}{100} \implies \iota = -2 + \frac{\sqrt{2}}{100},$$

an irrational number on **every** off-axis surface.

This achieves exact solenoidality, exact tangency to the original nested tori, non-axisymmetry, and irrational winding. **But it is not an MHD equilibrium.**

Force balance already fails on the axis

Let

$$\mathbf{h} = -v\mathbf{r}_u + \bar{u}\mathbf{r}_v.$$

On the magnetic axis, $\bar{u} = v = 0$, so $\mathbf{h} = 0$. Hence the trial field has exactly the original field and original field-line acceleration on the axis. Nevertheless, its changed transverse derivatives change the current.

Using $\tilde{\mathbf{B}}_\nu = \mathbf{B} + \nu\mathbf{h}$, one obtains exactly

$$\left[(\nabla \times \tilde{\mathbf{B}}_\nu) \times \tilde{\mathbf{B}}_\nu \right]_{\text{axis}} = -\nu \nabla (\mathbf{B} \cdot \mathbf{h})_{\text{axis}}. \quad (11)$$

At the axis point $t = 0$, define

$$k = \sqrt{1 - \epsilon^2}, \quad L_* = \sqrt{\frac{1+k}{2}}, \quad b = \sqrt{1 - \epsilon}.$$

The embedding gives

$$\mathbf{r}_t = (0, k, \epsilon), \quad \mathbf{r}_v = (0, b/L_*, 1), \quad \mathbf{r}_t \cdot \mathbf{r}_u = 0.$$

Therefore

$$\left[(\nabla \times \tilde{\mathbf{B}}_\nu) \times \tilde{\mathbf{B}}_\nu \right]_{\gamma(0)} = -\nu \left(\frac{kb}{L_*} + \epsilon \right) \nabla u \neq 0 \quad (12)$$

for $0 < \epsilon < 1$ and $\nu \neq 0$.

For irrational ν , any equilibrium pressure would have to be constant on each torus because the field lines are dense on it. Smoothness would then make the magnetic axis a critical set of pressure. Equation (12) is inconsistent with that.

For the concrete parameters

$$\epsilon = \frac{1}{4}, \quad \nu = \frac{\sqrt{2}}{100},$$

the force at $\gamma(0)$ is approximately

$$(-0.011998, 0, 0),$$

not a roundoff-level error.

5. What these calculations establish

The useful outcome is a fairly sharp separation of the available constructions:

Construction	Non-axisymmetric	Noninteger transform	Exact scalar-pressure equilibrium
Attached solution	Yes	No	Yes
Axisymmetric branch (3), $C = 2$	No	Yes	Yes
Unequally stretched branch (5), $C = 2$	Yes	Yes	No: obstruction (6)
Label-rotation field (9), irrational ν	Yes	Yes	No: axis force (12)

The verification script checks the successful and failed branches separately. On 100 sample points for each exact branch, the force-balance residual was below 1.4×10^{-15} . The failed stretched branch had a sampled curl-of-force magnitude as large as 1.49; the irrational label-rotation trial reproduced the nonzero axis force above.

I also checked whether published expansion constructions supplied the missing exact completion. Weitzner’s low-shear construction explicitly does not prove convergence, while the arbitrary-order near-axis construction is an asymptotic framework. Neither supplies the exact, finite-volume solution needed to complete the arguments here. [AIP Publishing +1](#)

The remaining mathematical requirement is a genuinely nonquadratic, non-affine construction that changes the winding while restoring scalar-pressure force balance. The quadratic-potential result proves that one major route is closed; it does not prove that your requested generalization is impossible. Neither of the non-axisymmetric trials above completes that generalization.

I found an exact analytic family that meets the requested conditions. In the concrete example below, every regular flux surface has

$$1 < \iota < 2,$$

and a specified regular surface has **exactly**

$\iota = \sqrt{2}.$

Moreover, the transform is an analytic, nonconstant function of the surface label, so **all but a countable set of the flux surfaces have irrational transform.**

This uses the same central idea as the attachment—construct a divergence-free field whose convective acceleration is a gradient—but a different ansatz. It is **not simply a detuning of the attachment’s formula**, and I have not shown that the two fields belong to a common continuously parameterized family. The attachment obtains gradient acceleration from oscillators with frequencies 1, 1, 2, which also locks the field lines into closed orbits. The construction below retains gradient acceleration without that period locking. [20260910-01 Astra analytic nona... +1](#)

All spatial dependence below is explicit in elementary functions. To obtain the particular value $\sqrt{2}$, one constant is fixed by a monotone scalar equation involving one-dimensional integrals—not by solving an equilibrium PDE.

1. The magnetic field, flux function, and pressure

Use dimensionless Cartesian coordinates and your normalization

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = \nabla p.$$

Let $\varepsilon, S, \kappa > 0$, and define

$$\zeta = x + iy, \quad \bar{\zeta} = x - iy,$$

$$K = \bar{\zeta} \sqrt{1 + \frac{\varepsilon}{\bar{\zeta}^2}}, \quad \Phi = \zeta K + \frac{\pi}{2} - S. \quad (1)$$

The square root in (1) is the principal square root of the expression **inside the parentheses**. We will work entirely in

$$x^2 + y^2 > \varepsilon,$$

where that expression has positive real part, so this defines a globally single-valued, real-analytic function.

The placement of the square root matters: replacing (1) by the principal square root of $\bar{\zeta}^2 + \varepsilon$ would introduce an artificial branch cut.

Define

$$w(x, y) = \frac{i\kappa}{2K} \sin \Phi.$$

The field is

$$\begin{aligned} B_x + iB_y &= e^{iz} w = \frac{i\kappa e^{iz}}{2K} \sin \Phi, \\ B_z &= -\kappa \operatorname{Re}(e^{iz} \cos \Phi). \end{aligned} \quad (2)$$

Thus the real and imaginary parts of the first expression give B_x and B_y .

The flux function and pressure are

$$\psi = \frac{1}{2} \left[\sin^2 z + \left(\frac{B_z}{\kappa} \right)^2 \right], \quad p = p_a - \kappa^2 \psi. \quad (3)$$

For a concrete example, choose

$$\varepsilon = \frac{1}{5}, \quad \kappa = 1, \quad p_a = 1, \quad \delta = \frac{1}{200}. \quad (4)$$

The remaining parameter is

$$S \approx 0.70435455152665. \quad (5)$$

Its **exact** definition, which makes $\iota = \sqrt{2}$ on $\psi = 1/400$, is given in section 5.

The domain will be an explicitly parameterized solid torus with $0 \leq \psi \leq \delta$. Restricting to this particular component is important: the formula (3), considered over a much larger region, has other components.

2. Exact verification of the MHD equations

Here is an analytic verification using ordinary Euclidean derivatives.

Solenoidality

Introduce the Wirtinger derivatives

$$\partial_\zeta = \frac{1}{2}(\partial_x - i\partial_y), \quad \partial_{\bar{\zeta}} = \frac{1}{2}(\partial_x + i\partial_y).$$

Since K depends only on $\bar{\zeta}$,

$$w_\zeta = \frac{i\kappa}{2} \cos \Phi.$$

Writing

$$D = 2w_\zeta,$$

equation (2) becomes

$$B_x + iB_y = e^{iz}w, \quad B_z = -\operatorname{Im}(e^{iz}D).$$

Consequently,

$$\partial_x B_x + \partial_y B_y = \operatorname{Re}(e^{iz}D),$$

whereas

$$\partial_z B_z = -\operatorname{Re}(e^{iz}D).$$

Therefore

$$\boxed{\nabla \cdot \mathbf{B} = 0.} \tag{6}$$

Gradient acceleration

The special choice of K gives

$$K^2 = \bar{\zeta}^2 + \varepsilon, \quad KK_{\bar{\zeta}} = \bar{\zeta}.$$

Direct differentiation then gives the identities

$$\boxed{w_{\zeta\zeta} = -K^2 w, \quad w_{\zeta\bar{\zeta}} = -|\zeta|^2 w, \quad ww_{\zeta\zeta} - w_\zeta^2 = \frac{\kappa^2}{4}.} \tag{7}$$

These identities are what make force balance work.

First, direct substitution into the horizontal convective derivative gives

$$\begin{aligned} [(\mathbf{B} \cdot \nabla)\mathbf{B}]_x + i[(\mathbf{B} \cdot \nabla)\mathbf{B}]_y &= \bar{w} w_\zeta + w \bar{w}_{\bar{\zeta}} \\ &= (\partial_x + i\partial_y) \frac{|w|^2}{2}. \end{aligned} \tag{8}$$

For the vertical component, put $E = e^{iz}D$. Equation (7) implies

$$wD_\zeta = \frac{D^2 + \kappa^2}{2}, \quad \bar{w}D_{\bar{\zeta}} = -2|\zeta|^2|w|^2 \in \mathbb{R}.$$

Hence

$$(\mathbf{B} \cdot \nabla)B_z = -\operatorname{Im} \left[\frac{E^2}{2} + \frac{\kappa^2 e^{2iz}}{2} - 2|\zeta|^2|w|^2 - i(\operatorname{Im} E)E \right].$$

But

$$\frac{E^2}{2} - i(\operatorname{Im} E)E = \frac{|E|^2}{2}$$

is real. It follows that

$$\boxed{(\mathbf{B} \cdot \nabla)B_z = -\kappa^2 \sin z \cos z.} \tag{9}$$

Combining (8)–(9),

$$\boxed{(\mathbf{B} \cdot \nabla) \mathbf{B} = -\nabla Q, \quad Q = -\frac{|w|^2}{2} + \frac{\kappa^2}{2} \sin^2 z.} \quad (10)$$

This is the analogue of the gradient-acceleration identity in the attachment. Here, however, the vertical field-line motion satisfies a pendulum equation rather than an oscillator with a fixed commensurate frequency.

Scalar-pressure force balance

Since

$$|\mathbf{B}|^2 = |w|^2 + B_z^2,$$

equations (3) and (10) imply

$$Q + \frac{|\mathbf{B}|^2}{2} = \frac{\kappa^2}{2} \sin^2 z + \frac{B_z^2}{2} = \kappa^2 \psi.$$

Therefore

$$\begin{aligned} (\nabla \times \mathbf{B}) \times \mathbf{B} &= (\mathbf{B} \cdot \nabla) \mathbf{B} - \nabla \frac{|\mathbf{B}|^2}{2} \\ &= -\kappa^2 \nabla \psi \\ &= \nabla p. \end{aligned}$$

Thus

$$\boxed{(\nabla \times \mathbf{B}) \times \mathbf{B} = \nabla p.} \quad (11)$$

Dotting (11) with $\mathbf{B}$ also establishes

$$\boxed{\mathbf{B} \cdot \nabla \psi = 0.} \quad (12)$$

No external force, pressure anisotropy, or non-Euclidean metric is being introduced. The quantity Q is only an auxiliary potential used to verify the identity.

3. The solid torus, nested surfaces, and magnetic axis

We can give an explicit global coordinate map, including the magnetic axis. This establishes the topology and the absence of other pressure-critical points.

Confocal coordinates in the xy -plane

For $\sigma > 0$, define

$$\begin{aligned} h(\sigma) &= \sqrt{4\sigma^2 + \varepsilon^2}, \\ a(\sigma) &= \sqrt{\frac{h(\sigma) - \varepsilon}{2}}, \quad b(\sigma) = \sqrt{\frac{h(\sigma) + \varepsilon}{2}}. \end{aligned}$$

In particular,

$$a(\sigma)b(\sigma) = \sigma, \quad b(\sigma)^2 - a(\sigma)^2 = \varepsilon.$$

Parameterize the horizontal coordinates by

$$x = a(\sigma) \cos t, \quad y = b(\sigma) \sin t, \quad t \in \mathbb{R}/(2\pi\mathbb{Z}). \quad (13)$$

These are nested confocal ellipses.

In these coordinates, the quantities in (1) simplify to

$$K = b(\sigma) \cos t - i a(\sigma) \sin t,$$

$$\Phi = \sigma + \frac{\pi}{2} - S + iV(t), \quad V(t) = \frac{\varepsilon}{2} \sin 2t. \quad (14)$$

Two nonsingular transverse labels

Introduce the real labels

$$P = -\frac{B_z}{\kappa}, \quad Z = \sin z. \quad (15)$$

Then

$$\psi = \frac{P^2 + Z^2}{2}.$$

Conversely, given P, Z, t in a small disk, define

$$\beta(Z, t) = \arctan \left(\frac{\tanh V(t) Z}{\sqrt{1 - Z^2}} \right),$$

and

$$\sigma(P, Z, t) = S + \beta(Z, t) - \arcsin \left(\frac{P}{\sqrt{\cosh^2 V(t) - Z^2}} \right). \quad (16)$$

The desired position map is

$$\mathbf{r}(P, Z, t) = (a(\sigma(P, Z, t)) \cos t, b(\sigma(P, Z, t)) \sin t, \arcsin Z). \quad (17)$$

Substitution into (2) verifies the inverse relations

$$-\frac{B_z(\mathbf{r}(P, Z, t))}{\kappa} = P, \quad \sin z(\mathbf{r}(P, Z, t)) = Z.$$

Therefore

$$\psi(\mathbf{r}(P, Z, t)) = \frac{P^2 + Z^2}{2}. \quad (18)$$

Define the physical domain by

$$\Omega_\delta = \{\mathbf{r}(P, Z, t) : P^2 + Z^2 \leq 2\delta, \quad t \in S^1\}. \quad (19)$$

Regularity and global injectivity

For the parameters (4), and any $S \in (3/5, 4/5)$, the disk in (19) has radius $1/10$. Equation (16) gives the uniform bound

$$|\sigma - S| \leq \arctan \left(\frac{\tanh(1/10)}{\sqrt{99}} \right) + \arcsin \left(\frac{1}{\sqrt{99}} \right) < 0.111.$$

Thus $\sigma > 0.489$, and the entire torus lies strictly inside the branch domain $x^2 + y^2 > \varepsilon$.

For an explicit Jacobian, define

$$G(\sigma, t) = b(\sigma)^2 \cos^2 t + a(\sigma)^2 \sin^2 t = \frac{h(\sigma) + \varepsilon \cos 2t}{2},$$

$$T(P, Z, t) = \sqrt{\cosh^2 V(t) - P^2 - Z^2}.$$

Then

$$\det \frac{\partial(x, y, z)}{\partial(P, Z, t)} = \frac{G(\sigma, t)}{h(\sigma) T(P, Z, t) \sqrt{1 - Z^2}} > 0. \quad (20)$$

The confocal coordinates (σ, t) are globally unique in this region. Together with $Z = \sin z$ and $P = -B_z/\kappa$, this also makes (17) globally injective.

It follows that Ω_δ is a **compact embedded solid torus**, and that every surface

$$0 < \psi = s \leq \delta$$

is the embedded torus

$$\mathbf{r} \left(-\sqrt{2s} \cos \chi, \sqrt{2s} \sin \chi, t \right), \quad (\chi, t) \in S^1 \times S^1. \quad (21)$$

The field, pressure, and flux function are real analytic on an open neighborhood of the entire solid torus.

Magnetic axis and pressure gradient

Equation (18) shows that the unique magnetic axis is $P = Z = 0$:

$$\gamma(t) = (a(S) \cos t, b(S) \sin t, 0). \quad (22)$$

For the calibrated example, its semiaxes are approximately

$$a(S) = 0.78193211585492, \quad b(S) = 0.90078734105523.$$

On the axis,

$$\mathbf{B}(\gamma(t)) = \frac{\kappa \cosh V(t)}{2G(S, t)} \gamma'(t),$$

so it is a nonvanishing closed field line.

Because (17) is a diffeomorphism and $\psi = (P^2 + Z^2)/2$,

$$\nabla \psi = 0 \iff P = Z = 0. \quad (23)$$

Consequently,

$$\nabla p \neq 0 \quad \text{everywhere off the magnetic axis.} \quad (24)$$

The example also has $p \geq 1 - \delta = 0.995 > 0$.

Genuine non-axisymmetry

For $\varepsilon > 0$, the axis (22) is an ellipse with unequal semiaxes, not a circle.

Any continuous Euclidean rotational symmetry of $\mathbf{B}$ would preserve the zero set of

$$(\nabla \times \mathbf{B}) \times \mathbf{B},$$

which in this torus is precisely that ellipse. A smooth closed curve invariant under a nontrivial continuous group of rotations must be a circle.

Therefore the field is **not axisymmetric about any straight line in $\mathbb{R}^3$** . The axis is planar, unlike the attachment's axis, but its noncircular shape is sufficient to establish genuine non-axisymmetry.

4. Exact rotational-transform formula

Fix a regular surface $\psi > 0$, and write

$$k = \sqrt{2\psi}, \quad P = -k \cos \chi, \quad Z = k \sin \chi.$$

Define

$$C(\chi) = \sqrt{1 - k^2 \sin^2 \chi}, \quad T(t) = \sqrt{\cosh^2 V(t) - k^2}.$$

Let a dot denote differentiation along a field line,

$$\frac{d\mathbf{r}}{d\tau} = \mathbf{B}.$$

Equations (9) and (15) give

$$\dot{P} = \kappa Z \sqrt{1 - Z^2}, \quad \dot{Z} = -\kappa P \sqrt{1 - Z^2}.$$

The angular equations are consequently

$$\dot{\chi} = \kappa C(\chi), \quad \dot{t} = \frac{\kappa T(t)}{h(\sigma(\chi, t)) + \varepsilon \cos 2t}. \quad (25)$$

Both are strictly positive throughout the chosen domain.

These angles count physical winding

The angle t counts ordinary toroidal turns: since $a, b > 0$, the physical cylindrical angle $\phi = \arg(x + iy)$ differs from t by a bounded periodic function.

The angle χ counts genuine poloidal turns, with the convention of increasing upward from the outboard side. At $\chi = 0$, we have $z = 0$ and $\sigma > S$, so the point is outside the axis ellipse. At $\chi = \pi$, it is inside that ellipse. For $0 < \chi < \pi$, $z > 0$, and for $\pi < \chi < 2\pi$, $z < 0$.

Thus χ has the usual meridional winding around the axis. **There is no added integer arising from a twisted choice of poloidal angle.**

Two explicit integrals

Define

$$\mathcal{A}_\varepsilon(k) = \int_0^{2\pi} \frac{dt}{\sqrt{\cosh^2\left(\frac{\varepsilon}{2} \sin 2t\right) - k^2}}, \quad (26)$$

and

$$\mathcal{C}_{\varepsilon, S}(k) = \int_0^{2\pi} \frac{d\chi}{h\left(S + \arctan \frac{k \cos \chi}{\sqrt{1 - k^2}}\right) \sqrt{1 - k^2 \sin^2 \chi}}. \quad (27)$$

The rotational transform is exactly

$$\iota(\psi) = \frac{\mathcal{A}_\varepsilon(\sqrt{2\psi})}{\mathcal{C}_{\varepsilon, S}(\sqrt{2\psi})}. \quad (28)$$

Here is a derivation that does not rely on numerical field-line following.

The absolute volume Jacobian in (ψ, χ, t) coordinates is

$$\mathcal{J} = \frac{G}{hTC}.$$

Solenoidality makes the following one-form on each flux torus closed:

$$\begin{aligned} \alpha &= \mathcal{J}(\dot{\chi} dt - \dot{t} d\chi) \\ &= \frac{\kappa}{2} \left[\frac{1 + \varepsilon \cos 2t/h}{T(t)} dt - \frac{1}{hC(\chi)} d\chi \right]. \end{aligned} \quad (29)$$

It also satisfies $\alpha(\mathbf{B}) = 0$.

Its period around the t -cycle is

$$\oint_t \alpha = \frac{\kappa}{2} \mathcal{A}_\varepsilon(k).$$

To see this, evaluate the cycle at $\chi = \pi/2$, where $P = 0$. The term involving $\varepsilon \cos 2t = V'(t)$ integrates to zero because its remaining coefficient is a single-valued function of V .

Its period around the χ -cycle is

$$\oint_\chi \alpha = -\frac{\kappa}{2} \mathcal{C}_{\varepsilon,S}(k).$$

Here one evaluates at $t = 0$, for which

$$\sigma = S + \arctan \frac{k \cos \chi}{\sqrt{1-k^2}}.$$

A closed one-form differs from the constant one-form with these periods by the differential of a periodic function. Since α annihilates the field, along any field line

$$\mathcal{A}_\varepsilon(k) \Delta t - \mathcal{C}_{\varepsilon,S}(k) \Delta \chi$$

remains bounded. Dividing by Δt proves (28).

In particular, this formula gives the same transform for every field line on a given surface.

5. An exactly irrational regular surface—and no integer surfaces

Choose the parameters in (4), and designate the regular surface

$$\psi_* = \frac{1}{400}, \quad k_* = \sqrt{2\psi_*} = \frac{1}{\sqrt{200}}.$$

Define S **exactly** as the unique root in $(3/5, 4/5)$ of

$$\frac{\mathcal{A}_{1/5}(1/\sqrt{200})}{\mathcal{C}_{1/5,S}(1/\sqrt{200})} = \sqrt{2}. \quad (30)$$

Then, by (28),

$$\iota(\psi_*) = \sqrt{2}. \quad (31)$$

This is an exact definition of a constant. The decimal value in (5) is only its numerical approximation.

Existence and uniqueness of the required constant

For fixed k , when $S > \arcsin k$, the integrand in (27) strictly decreases with S . Hence $\mathcal{C}_{\varepsilon,S}(k)$ strictly decreases and ι strictly increases with S .

Useful elementary bounds are obtained by defining

$$h_\pm = \sqrt{4(S \pm \arcsin k)^2 + \varepsilon^2}.$$

Then

$$h_- \frac{\sqrt{1-k^2}}{\sqrt{\cosh^2(\varepsilon/2) - k^2}} \leq \iota \leq \frac{h_+}{\sqrt{1-k^2}}. \quad (32)$$

These follow directly by bounding the integrands in (26)–(27).

At $k = k_*$, these inequalities give

$$\iota|_{S=3/5} < 1.360 < \sqrt{2},$$

$$\iota|_{S=4/5} > 1.464 > \sqrt{2}.$$

Continuity and strict monotonicity establish existence and uniqueness of the root in (30).

Every regular surface has noninteger transform

Throughout the selected solid torus,

$$0 \leq k \leq \frac{1}{10}, \quad \frac{3}{5} < S < \frac{4}{5}.$$

The same bounds (32), uniformly over these intervals, give

$$1.0143 \dots \leq \iota(\psi) \leq 1.8206 \dots$$

In particular,

$$\boxed{1 < \iota(\psi) < 2 \quad (0 < \psi \leq \delta).} \quad (33)$$

Therefore the one-toroidal-turn return map is not the identity on any regular surface. On the surface $\psi = \psi_*$, no field line closes.

Almost every surface has irrational transform

The integral formula (28) is analytic in ψ , including at $\psi = 0$. It is also nonconstant; this can be established analytically rather than inferred from numerical values.

Define

$$M_j = \frac{1}{2\pi} \int_0^{2\pi} \operatorname{sech}^j \left(\frac{\varepsilon}{2} \sin 2t \right) dt.$$

Expanding (26)–(27) at the axis gives

$$\iota(0) = h(S)M_1$$

and

$$\boxed{\frac{\iota'(0)}{\iota(0)} = \frac{M_3}{M_1} - \frac{1}{2} \left[1 + \frac{32S^2 - 4\varepsilon^2}{(4S^2 + \varepsilon^2)^2} \right]}. \quad (34)$$

For $\varepsilon = 1/5$ and $S \in (3/5, 4/5)$,

$$\frac{32S^2 - 4\varepsilon^2}{(4S^2 + \varepsilon^2)^2} > 3, \quad \frac{M_3}{M_1} \leq 1.$$

Thus $\iota'(0) < 0$, proving nonconstancy.

For a nonconstant analytic function, the preimage of each particular rational number is discrete. Since the rationals are countable, the flux surfaces with rational transform form a countable set. Hence **all but a countable set of the regular surfaces have irrational transform**.

There are also rational surfaces, as is unavoidable for a continuous nonconstant transform profile, but none have integer transform in this domain.

6. Numerical values and reproducible checks

For the parameter defined by (30), evaluation of the exact integral formula gives:

ψ	$\iota(\psi)$
0, interpreted as the axis limit	1.41928965680794

ψ	$\iota(\psi)$
0.001	1.41726635231145
1/400	$\sqrt{2} = 1.414213562373095 \dots$
1/200, the boundary	1.40907746396421

I also checked the Cartesian expressions using 64-bit automatic differentiation at 1,202 points, including points on the axis and boundary. The maximum residuals were

$$\begin{aligned} |\nabla \cdot \mathbf{B}| &< 3.5 \times 10^{-16}, \\ |(\nabla \times \mathbf{B}) \times \mathbf{B} - \nabla p| &< 3.9 \times 10^{-16}, \\ |\mathbf{B} \cdot \nabla \psi| &< 2.1 \times 10^{-17}. \end{aligned}$$

Independent fourth-order finite-difference checks also corroborated the equations. These checks supplement the analytic identities; they are not the basis for the irrationality assertion.

The implementation includes the Cartesian field, pressure and flux function, explicit surface embedding, scalar parameter calibration, transform integrals, and differential checks:

[`</>` Download the Python implementation and verification script](#)

[`</>` Download the numerical verification results](#)

Thus this construction supplies a full analytic nested-torus equilibrium—not merely isolated surviving tori—with smooth scalar pressure, nonzero pressure gradient off its elliptical magnetic axis, genuine non-axisymmetry, and noninteger transform throughout the torus. The specified regular surface has exactly irrational transform, and irrational surfaces are the rule rather than the exception. I have not established whether this particular analytic family has appeared previously in the literature.



mhd_generalization_research_note.md

Document



mhd_generalization_checks.py

Code



exact_nonaxisymmetric_mhd_irrational_transform.py

Code



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 Sources

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